LOCAL QUASI-PROJECTIVITY IS LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- H-quasi-projective morphisms are locally quasi-projective. Lemma \ref{lemma-H-quasi-projective-quasi-projective}, cited at proof lines 9799-9800. Gives one implication.
- On an affine base a relatively ample sheaf can be replaced by an ample one. Lemma \ref{lemma-finite-type-over-affine-ample-very-ample}, cited at proof lines 9805-9806. Produces an ample invertible sheaf.
- The existence of an immersion into a projective space over the base. Lemma \ref{lemma-quasi-projective-finite-type-over-S}, cited at proof lines 9807-9809. Gives H-quasi-projectivity.

Source reference. Original lemma 01VZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2464–2464. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01VY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2463–2463. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01VT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2458–2458. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01VS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2457–2457. Navigation pointer to a statement the

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent:*

(1) *The morphism f is locally quasi-projective.*
(2) *There exists an open covering $S = \bigcup V_j$ such that each $f^{-1}(V_j) \to V_j$ is H-quasi-projective.*

Proof. By Lemma 01VY we see that (2) implies (1). Assume (1). The question is local on S and hence we may assume S is affine, X of finite type over S and $\mathcal{L}$ is a relatively ample invertible sheaf on X/S. By Lemma 01VT we may assume $\mathcal{L}$ is ample on X. By Lemma 01VS we see that there exists an immersion of X into a projective space over S, i.e., X is H-quasi-projective over S as desired. □